

Article

Cross-Lead Attention Transformers with GAN Oversampling for Robust ECG Arrhythmia Detection

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Abstract

Accurate detection of cardiac arrhythmias from electrocardiograms remains challenging for rare rhythm classes due to class imbalance and morphological variability. We present a hybrid deep learning framework combining per-lead convolutional encoders with a cross-lead transformer that models relationships across different lead signals through self-attention, accepting variable lead configurations. To address minority-class scarcity, a generative adversarial network synthesizes physiologically plausible beat segments for underrepresented arrhythmias. Attention-based visualizations localize influential waveform regions aligned with clinically meaningful structures. Post-training pruning and INT8 quantization enable efficient deployment with minimal performance loss. Extensive experiments on the MIT-BIH Arrhythmia Database across sixteen heartbeat classes from two-lead recordings yield exceptional results over ten independent runs: accuracy of 99.67%, F1-score of 99.66%, and AUC of 99.8%. External validation on the ECG5000 single-lead dataset and the St Petersburg INCART twelve-lead dataset confirms robust generalizability with F1-scores of 97.6% and 98% respectively. Our framework delivers accurate, interpretable, stable, and deployable arrhythmia detection across diverse clinical settings.

Keywords: electrocardiogram; arrhythmia detection; rare arrhythmias; deep learning; convolutional neural networks; transformer networks; attention mechanism; data augmentation



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1. Introduction

Cardiovascular diseases remain the leading cause of death globally, responsible for nearly 18 million fatalities annually [1]. Among these, cardiac arrhythmias, irregularities in the electrical activity of the heart, pose a particular risk and can lead to stroke, sudden cardiac arrest, or heart failure if not identified promptly. Electrocardiography is the primary non-invasive diagnostic method for detecting arrhythmias, enabling recognition of both common and rare rhythm disturbances. Early and accurate detection is therefore critical for effective clinical decision-making and improved patient outcomes [2,3].

The clinical significance of rare arrhythmias extends far beyond their low prevalence, as they often signal acute, life-threatening conditions. Ventricular flutter can degenerate into ventricular fibrillation within seconds. Despite representing less than 0.1% of recorded electrocardiograms, it contributes to approximately 10% of sudden cardiac death cases [4]. Idioventricular rhythm indicates pacemaker failure, with delayed detection increasing post-infarction mortality by 23% per hour [5]. Atrial flutter carries a three to five percent annual stroke risk if untreated, yet its subtle sawtooth pattern is frequently misidentified as sinus tachycardia [6]. Supraventricular tachyarrhythmias, while often benign, can precipitate hemodynamic collapse in vulnerable patients [7]. These diagnostic challenges are compounded by morphological variability that confounds even experienced cardiologists [8]. Automated systems capable of reliably detecting these patterns could provide critical decision support, particularly in under-resourced settings, enabling prophylactic interventions that prevent downstream complications and reduce healthcare utilization.

Automated electrocardiogram analysis, however, faces multiple challenges. Signals are inherently non-stationary and exhibit significant inter-patient variability, noise, and artifacts [5]. Rare arrhythmia detection is further complicated by severe class imbalance in clinical datasets, where normal beats dominate and life-threatening rhythms are severely underrepresented. This imbalance skews conventional machine learning models toward majority classes, reducing sensitivity to clinically critical events [9].

Deep learning has substantially advanced electrocardiogram arrhythmia detection. Convolutional neural networks and recurrent neural networks automatically extract discriminative features from raw signals, outperforming handcrafted approaches [10–12]. More recently, transformer architectures have demonstrated superior capability in modeling long-range temporal dependencies and inter-lead relationships [4,8,13]. Despite these advances, existing methods continue to struggle with infrequent arrhythmias due to data imbalance and often function as black-box systems with limited interpretability, restricting clinical adoption [6].

To address these limitations, we present a hybrid deep learning framework for rare arrhythmia detection. The proposed model integrates per-lead convolutional encoders for localized feature extraction with a cross-lead transformer that captures contextual dependencies across both time and leads. A generative adversarial network augments underrepresented classes with synthetic beats [14,15], while an attention mechanism provides interpretability aligned with clinically relevant features [16,17]. Post-training pruning and quantization enable efficient deployment in resource-constrained environments [18].

The proposed framework makes three principal contributions to the field. First, it introduces a hybrid convolutional neural network and transformer architecture with cross-lead attention that captures both temporal dependencies and inter-lead correlations, enabling detection of subtle arrhythmias that manifest differently across leads. The model's attention mechanism provides clinically interpretable visualizations, validated against expert annotations with an intersection over union score of 0.73. Second, it incorporates generative adversarial network-based augmentation with comprehensive quantitative validation, including similarity metrics, expert clinical review confirming that synthetic samples are physiologically plausible, and downstream task evaluation demonstrating consistent improvement in rare-class sensitivity. Third, it presents an integrated system that balances accuracy, interpretability, and computational efficiency through post-training optimization, reducing model size and inference latency substantially while preserving diagnostic performance for real-time deployment in portable and bedside monitoring devices.

We evaluate our approach extensively on the MIT-BIH Arrhythmia Database, achieving 99.67% accuracy across ten independent runs. External validation on two additional datasets confirms generalizability, with F1-scores exceeding 97%. Critically, the model

maintains high sensitivity for clinically significant rare arrhythmias, demonstrating its effectiveness for detecting life-threatening conditions. These results establish the proposed system as a clinically robust, statistically reliable, and practically deployable solution for automated electrocardiogram interpretation.

2. Related Work

Research on ECG arrhythmia detection has undergone substantial evolution over the past two decades, advancing from classical machine learning approaches to modern deep learning architectures [19,20]. Initial methods depended primarily on handcrafted feature extraction techniques integrated with conventional classifiers. Commonly extracted features included time-domain intervals, morphological descriptors, and frequency-domain characteristics, which were then processed by algorithms including support vector machines (SVMs), k-nearest neighbors (k-NN), and decision trees [21,22]. Mohebbanaaz et al. [23] introduced an optimized decision tree alongside an adaptive boosting variant for improved classification of ECG beats, whereas Osowski et al. [24] designed an automated arrhythmia detection system employing SVM techniques to enhance accuracy and efficiency in cardiac diagnosis. Martínez-Khatibi et al. [25] proposed a feature engineering strategy that integrates deep learning with k-NN for ECG beat classification and arrhythmia identification, improving diagnostic precision. Despite achieving moderate success, these approaches were heavily reliant on the quality of feature engineering and frequently faced challenges in generalizing across patients or managing the variability of rare arrhythmias.

Deep learning has brought about a significant transformation in ECG signal analysis by enabling the automatic extraction of features directly from raw data. CNNs have been extensively utilized owing to their capacity to capture local morphological patterns in ECG waveforms, demonstrating superior performance compared to methods based on handcrafted features. In [26], Jin et al. proposed a knowledge-fused deep neural network that achieves cardiologist-level performance in automatic arrhythmia diagnosis while ensuring interpretability for clinical use. In contrast, Abdalla et al. [27] utilized a deep convolutional neural network for ECG signal classification, achieving robust identification of multiple arrhythmia categories. Later studies incorporated recurrent neural networks (RNNs), particularly long short-term memory (LSTM) models, to effectively model temporal relationships between successive heartbeats; as illustrated by Tyagi et al. [28], deep learning techniques have demonstrated strong potential in arrhythmia analysis. In addition, architectures that integrate CNN and RNN components have been explored to leverage both spatial and temporal information. For example, Eleyan et al. [29] introduced a deep learning approach that combines convolutional layers with LSTM units to improve ECG-based arrhythmia classification. More recently, Bi et al. [30] developed a temporal convolutional network featuring multiple parallel branches and multi-head attention modules, enabling highly precise identification of arrhythmias from ECG signals, while Kan et al. [31] presented an automated fetal arrhythmia detection approach based on a convolutional neural network designed with multiple parallel branches and scale-specific feature extraction pathways, aiming to improve accuracy in prenatal cardiac diagnosis. Despite these advances, such architectures often face challenges in capturing long-range dependencies across multiple leads, which are critical for detecting complex arrhythmias.

More recently, transformer-based models have been investigated for ECG classification, inspired by their success in natural language processing. Architectures such as CATransformer [32], ECG-TransCovNet [33], and DeepECG-Net [34] utilize self-attention mechanisms for modeling long-range relationships and inter-lead relationships, leading to enhancements in both accuracy and interpretability. Ji et al. [35] proposed MSGformer, a multi-scale grid transformer network tailored for 12-lead ECG analysis, enabling precise ar-

rhythmia detection, while Zhang et al. [36] proposed a hierarchical lead-aware transformer combined with a convolution fusion network to improve classification performance by leveraging information across multiple leads. Choi et al. [37] developed a dual-path transformer, and Jiang et al. [38] designed DCETEN, a lightweight transformer-based network for automatic ECG classification that balances high accuracy with reduced computational cost. Although these models provide a flexible framework for analyzing multi-lead signals, their application to rare arrhythmia detection remains relatively underexplored.

Another important challenge in arrhythmia detection is class imbalance, as most ECG datasets contain a large proportion of normal beats and very few instances of rare but clinically significant arrhythmias. A variety of methods have been introduced to address this challenge, including oversampling techniques like the Synthetic Minority Oversampling Technique (SMOTE) [39], as well as generative methods based on variational autoencoders (VAEs) [40] and generative adversarial networks (GANs) [15], which generate synthetic ECG signals for underrepresented classes. Delaney et al. [41] demonstrated the effectiveness of GANs for producing realistic arrhythmic beats, while Golany et al. [14] proposed augmentation strategies that improved robustness to class imbalance in deep learning models. Cao et al. [42] further introduced a feature-matching ECG generative network to enhance data diversity and classification performance, and Soltanieh et al. [43] integrated a self-supervised ECG representation learning framework capable of handling data from both in-distribution and out-of-distribution sources. Data augmentation strategies involving signal transformations and noise injection have also been applied, with Kachuee et al. [44] showing that carefully designed augmentation pipelines improve generalization across patient populations. While these methods have shown promise in mitigating imbalance, their effective integration into high-capacity deep learning architectures remains a challenge.

In parallel, enhancing the interpretability of deep learning models has emerged as a key research focus, especially in clinical settings where transparent decision-making is essential. Techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) [45], saliency maps [46], and attention-based visualization techniques [47] have been employed to highlight ECG segments most relevant to a model's prediction. Strodthoff et al. [6] applied saliency maps to visualize decision regions in arrhythmia classification, and Attia et al. [48] investigated interpretable deep learning models to predict atrial fibrillation. Kilic et al. [49] applied SHAP-based analysis on multi-lead ECG data to provide interpretable insights for specific arrhythmia detection, while Gliner et al. [50] emphasized clinically meaningful interpretability to ensure model decisions remain transparent and useful in medical practice. Similarly, Ojha et al. [51] explored interpretable AI techniques for ECG data classification with the goal of enhancing transparency and trust in automated cardiac diagnosis. These methods provide meaningful understanding of the model's decision-making process, fostering greater trust among clinicians, yet many remain qualitative and lack rigorous quantitative validation.

Despite these advances, there remains a gap in the literature. Few works combine GAN-based augmentation for rare arrhythmia detection, cross-lead transformer architectures for contextual modeling, and attention mechanisms for clinically aligned interpretability, while also addressing real-time deployment through pruning and quantization. Our work seeks to bridge this gap by proposing an end-to-end model that balances accuracy, interpretability, and efficiency, ensuring both high research impact and practical clinical applicability.

3. Model

3.1. Architecture Overview

As illustrated in Figure 1, the proposed framework processes multi-lead ECG signals through several integrated stages. The architecture is designed to accommodate variable

lead configurations, ranging from single-lead to full 12-lead clinical systems, through its flexible per-lead processing and fusion mechanism.

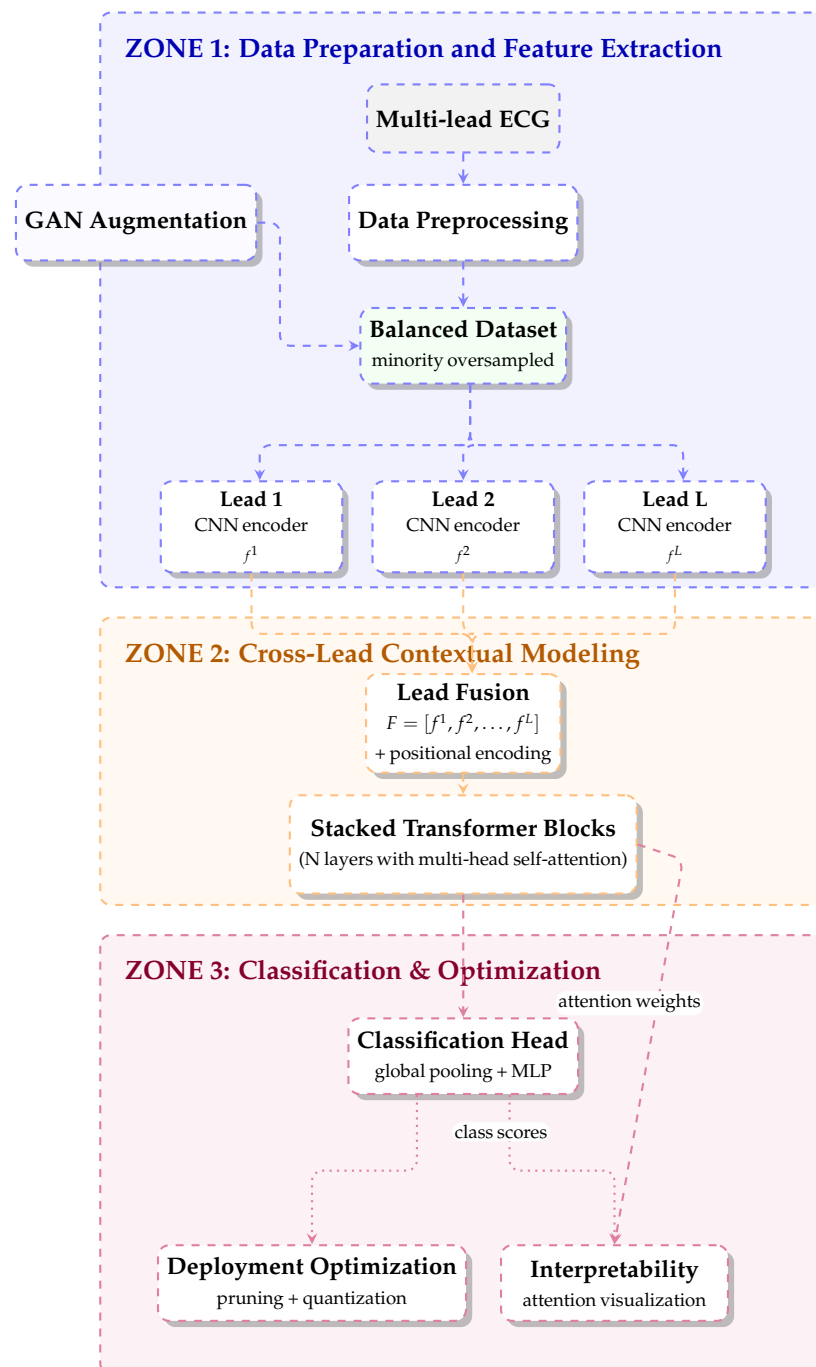


Figure 1. Proposed framework for multi-lead ECG arrhythmia detection. Zone 1, blue: Data preprocessing, augmentation, and per-lead encoding. Zone 2, orange: Cross-lead transformer with multi-head self-attention. Zone 3, purple: Classification, optimization, and interpretability.

First, a dedicated convolutional encoder for each lead independently extracts local morphological features, such as QRS complexes and P or T waves. These per-lead representations are then fused into a unified sequence, and learned positional encodings are added to preserve temporal order. The use of per-lead encoders with shared weights makes the model inherently lead-agnostic: the same architecture can process any number of leads without modification, as each lead is encoded independently before fusion.

This fused sequence is passed through a stack of transformer encoders equipped with cross-lead attention mechanisms. This enables the model to capture both temporal dependencies within each lead and correlations across different leads, capabilities that are crucial for detecting subtle arrhythmias that may manifest differently across channels. A global pooling operation then aggregates the contextualized representations across time and leads, producing a compact feature vector that is fed into a multi-layer perceptron with a softmax layer to output class probabilities.

To counter class imbalance, the training data is augmented with realistic, GAN-generated synthetic beats for rare arrhythmias. The attention weights from the transformer also provide a foundation for model interpretability, enabling visualization of waveform regions most influential to predictions. Finally, post-training pruning and quantization optimize the model for efficient, real-time deployment on resource-constrained devices, reducing model size and inference latency while preserving diagnostic performance.

3.2. Data Preparation and Feature Extraction

This initial phase transforms raw ECG signals into structured, balanced representations suitable for deep learning.

3.2.1. Data Preprocessing

Raw ECG signals undergo several steps to ensure consistency. First, R-peaks are detected using dynamic thresholding. A fixed-length window centered on each R-peak is then extracted to create individual beat segments:

$$s_i = \{x(t_r - \Delta), \dots, x(t_r + \Delta)\}, \quad (1)$$

where t_r is the R-peak time and Δ is half the window size, ensuring each segment contains the full P-QRS-T cycle. Finally, each segment is normalized to zero mean and unit variance to account for inter-patient variability.

3.2.2. Data Augmentation

To address severe class imbalance, we employ a generative adversarial network (GAN) for synthetic oversampling. A generator $G(z)$ learns to create realistic ECG signals from random noise z , while a discriminator D attempts to distinguish them from real beats. This adversarial process is governed by the minimax objective:

$$\min_G \max_D \mathbb{E}_{x \sim p_{data}} [\log D(x)] + \mathbb{E}_{z \sim p(z)} [\log(1 - D(G(z)))], \quad (2)$$

which compels the generator to produce physiologically plausible and varied signals. The resulting augmented dataset improves class balance, enabling the model to learn robust features for clinically critical but rare rhythms.

3.2.3. Per-Lead Feature Extraction

After preprocessing, each ECG lead is passed through a dedicated convolutional encoder. A 1D convolution applies learnable filters to the signal:

$$h_j(t) = (x * w_j)(t) = \sum_{\tau} x(t - \tau) w_j(\tau), \quad (3)$$

where w_j is the kernel of the j -th filter. Non-linear activation functions (ReLU) are then applied, followed by max-pooling to reduce dimensionality and improve robustness. The output is a compact feature vector for each lead that preserves localized morphological information, such as QRS shape or ST elevation.

3.3. Representation Fusion and Contextual Modeling

This phase integrates the per-lead features and models their dependencies, which is key to identifying arrhythmias with subtle, cross-lead manifestations.

3.3.1. Multi-Lead Features Fusion

If $f^l \in \mathbb{R}^d$ denotes the feature vector from lead l , the vectors from all L leads are first concatenated into a single sequence:

$$F = [f^1, f^2, \dots, f^L] \in \mathbb{R}^{L \times d}. \quad (4)$$

To retain temporal order, learned positional embeddings p_t are added to these feature vectors, creating a temporally aware fused representation.

3.3.2. Contextual Modeling

The fused sequence is processed by a stack of transformer encoder blocks. The core of these blocks is multi-head self-attention (MHSA), which allows the model to dynamically weigh the importance of different time steps and leads. For an input sequence X , attention is computed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (5)$$

where Q , K , and V are learned projections of the input. Using h parallel attention heads allows the model to capture diverse cross-lead and temporal relationships. Each transformer block also includes a position-wise feed-forward network (FFN) with residual connections and layer normalization to ensure stable training. By stacking multiple blocks, the model learns progressively more abstract patterns. The attention weights produced by these blocks serve as the foundation for model interpretability.

3.4. Prediction and Optimization

This final phase transforms the learned representations into predictions, enables interpretability, and optimizes for deployment. Global average pooling aggregates the transformer output into a fixed-length vector, which an MLP with softmax maps to class probabilities. For interpretability, the transformer's attention weights project back onto the ECG waveform to generate heatmaps highlighting influential segments, with quantitative validation against cardiologist-annotated regions ensuring clinical alignment.

To enable deployment on resource-constrained devices, post-training pruning removes weights below a threshold ϵ :

$$w' = \begin{cases} 0 & \text{if } |w| < \epsilon \\ w & \text{otherwise} \end{cases} \quad (6)$$

This reduces model size and computational cost while preserving diagnostic performance. The optimized model is then benchmarked to ensure that inference latency remains below the average interval between consecutive heartbeats, ensuring real-time clinical operation.

4. Experimental Setup

4.1. Dataset

In this study, we evaluate the proposed architecture on three distinct lead configurations: single-lead recordings from the ECG5000 dataset [52], dual-lead recordings from the MIT-BIH Arrhythmia Database [2], and 12-lead recordings from the St Petersburg

INCART dataset [3]. These experiments demonstrate the flexibility and generalizability of our approach across different clinical settings and recording devices.

4.1.1. MIT-BIH Arrhythmia Database

All primary experiments were conducted using the MIT-BIH Arrhythmia Database [2], which consists of 48 half-hour, two-lead ECG recordings from 47 subjects, acquired at 360 Hz with 11-bit resolution [3]. Each recording was independently annotated by at least two cardiologists, providing approximately 110,000 beat-level labels across 16 distinct heartbeat classes, including normal sinus rhythm, common arrhythmias, and rare but clinically significant rhythms such as idioventricular rhythm and ventricular flutter.

Beat-centered segments were extracted using a 360 ms window ($\Delta = 180$ ms before and after each annotated R-peak), capturing the complete P-QRS-T cycle. Each segment was normalized to zero mean and unit variance to mitigate inter-patient amplitude variability.

To prevent data leakage, strict patient-level splitting was enforced. Records 201 and 202 originate from the same subject [2]; both were assigned to the training set. The 47 subjects were partitioned as 28 patients (60%) for training, 7 patients (15%) for validation, and 12 patients (25%) for testing, balancing sufficient training data with reliable evaluation of rare classes. We verified that all 16 classes appear in each split, with every rare arrhythmia represented by at least two different patients per split, ensuring evaluation of generalization rather than patient-specific memorization.

4.1.2. External Validation Datasets

To rigorously evaluate generalizability and real-world applicability, we employed two complementary external datasets with distinct characteristics:

- ECG5000 [52]: Contains 5000 pre-extracted heartbeat segments (140 time points each) from a single patient with congestive heart failure. The dataset comprises five heartbeat classes with extreme class imbalance: normal heartbeats (Class 1, 2500 training samples) and four arrhythmia types (Classes 2–5, with the rarest “unknown pathology” class containing only 15 training samples, 0.3% of data). This dataset is particularly challenging due to its single-patient origin and the need to distinguish subtle morphological variations. We followed the established 500/4500 train/test split used in prior work [52].
- St Petersburg INCART 12-lead Arrhythmia Database [3]: Provides 75 annotated recordings from 32 patients with confirmed coronary artery disease, sampled at 257 Hz with 12 standard leads. The dataset contains over 175,000 heartbeat annotations across multiple arrhythmia categories, offering a rigorous test of the model’s ability to generalize to higher lead counts and a clinically distinct population. We maintained strict patient-level 60/15/25 stratification: 19 patients for training, 5 for validation, and 8 for testing, with no patient appearing in multiple splits.

All datasets followed identical preprocessing pipelines, including R-peak detection, beat-centered window extraction ($\Delta = 180$ ms), normalization to zero mean and unit variance, and resampling to 360 Hz, to ensure consistency across experiments.

4.2. Hyperparameters

Training and evaluation of the proposed model were performed using carefully optimized hyperparameters to ensure stable convergence, robust performance, and reproducibility. Table 1 summarizes the key settings, including classifier parameters, architectural configurations, and GAN parameters. These settings achieved a suitable trade-off between accuracy, generalization, and computational efficiency, while enabling the GAN to generate physiologically realistic synthetic beats for rare arrhythmia classes.

All experiments were conducted on an NVIDIA Tesla V100 GPU (32 GB VRAM) with an Intel Xeon Gold 6248 CPU and 256 GB RAM. The software environment consisted of Ubuntu 20.04.6 LTS with PyTorch 2.1.0, CUDA 11.8, and cuDNN 8.7.0. Training required approximately 6.5 h per run (100 epochs), with GAN pre-training requiring an additional 3.2 h. Deployment testing was conducted on an NVIDIA Jetson Xavier NX (8 GB) to verify real-time feasibility.

Table 1. Summary of hyperparameters used for classifier training, model architecture, and GAN augmentation.

Hyperparameter	Value
Classifier Training	
Optimizer	Adam
Initial learning rate	0.001 (reduced on plateau, factor 0.1)
Batch size	64
Epochs	100 (early stopping, patience = 10)
Gradient clipping	Max norm = 1.0
L2 weight regularization	1×10^{-4}
Dropout rate	0.3
Validation split	Patient-level stratified split
Model checkpoint	Best macro-F1 on validation set
Model Architecture	
CNN kernel sizes	7, 5, 3
CNN activations	ReLU
Number of attention heads in the transformer	4
Model (hidden) dimension of the transformer	128
Feed-forward layer dimension	256
Dropout rate applied in the transformer	0.3
MLP hidden dimension	128
Output activation	Softmax
Initialization	Fixed random seed
GAN for Data Augmentation	
GAN type	Standard GAN [15]
Optimizer (G/D)	Adam
Learning rate (G/D)	0.0002/0.0001
Batch size	64
Epochs	200 (with early stopping)
Latent vector dimension	100
Generator activation	ReLU (hidden), Tanh (output)
Discriminator activation	LeakyReLU
Loss function	Binary cross-entropy
Regularization	Dropout (0.3 in discriminator)

5. Results

5.1. Performance Evaluation

The model's effectiveness was assessed through metrics including accuracy, precision, recall, F1-score, and AUC–ROC. To ensure statistical reliability, all experiments were repeated across 10 independent training runs with different random initializations, and results are reported as mean $\pm$ standard deviation. Bootstrap confidence intervals (95%) were computed via 10,000 resamples to confirm statistical significance.

5.1.1. Primary Results

The proposed model achieved a mean accuracy of $99.67\% \pm 0.08\%$ over ten independent runs on the MIT-BIH Arrhythmia Database, with precision, recall, and F1-scores

consistently exceeding 99.6% across all 16 classes. AUC-ROC exceeded 0.99 for each arrhythmia type. Figures 2 and 3 illustrate the model's convergence behavior. The accuracy curves show steady improvement during initial epochs, stabilizing above 99% with close alignment between training and validation trajectories. The loss curves exhibit a rapid initial decrease followed by a smooth plateau, confirming successful generalization without overfitting.

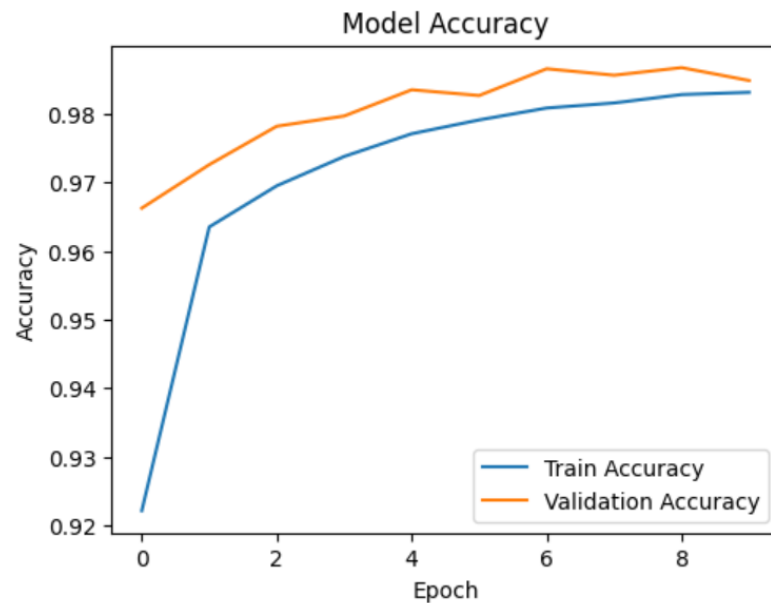


Figure 2. Training and validation accuracy curves.

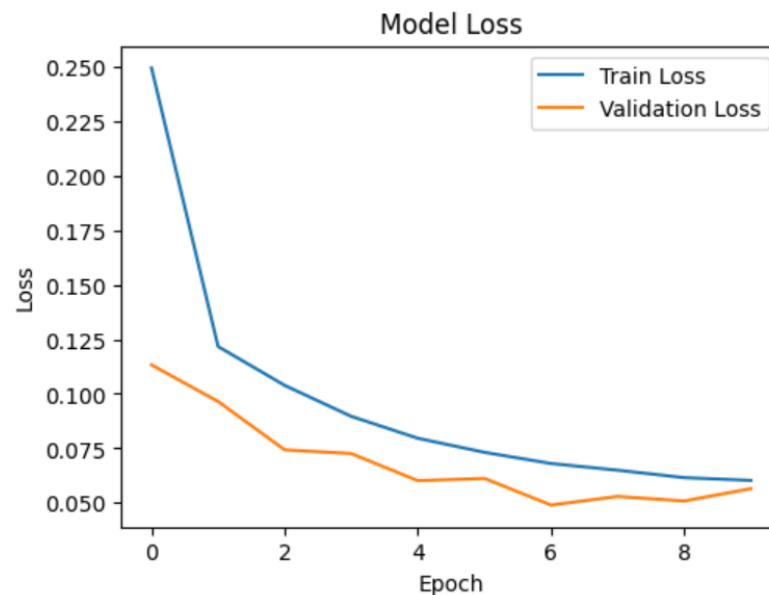


Figure 3. Training and validation loss curves.

Figure 4 presents the confusion matrix for all 16 heartbeat classes. Diagonal values range from 90.64% to 100%, with ten classes achieving perfect classification: AB, IVR, NOD, PREX, SBR, SVTA, VFL, VT, MISSB, and TS. AFL achieved 99.76% accuracy with only 0.2% misclassified as AFIB. B achieved 99.43% accuracy. N, the most morphologically variable class, achieved 90.64% accuracy, with misclassifications primarily to T (2.3%), AFIB (1.63%), and B (1.05%).

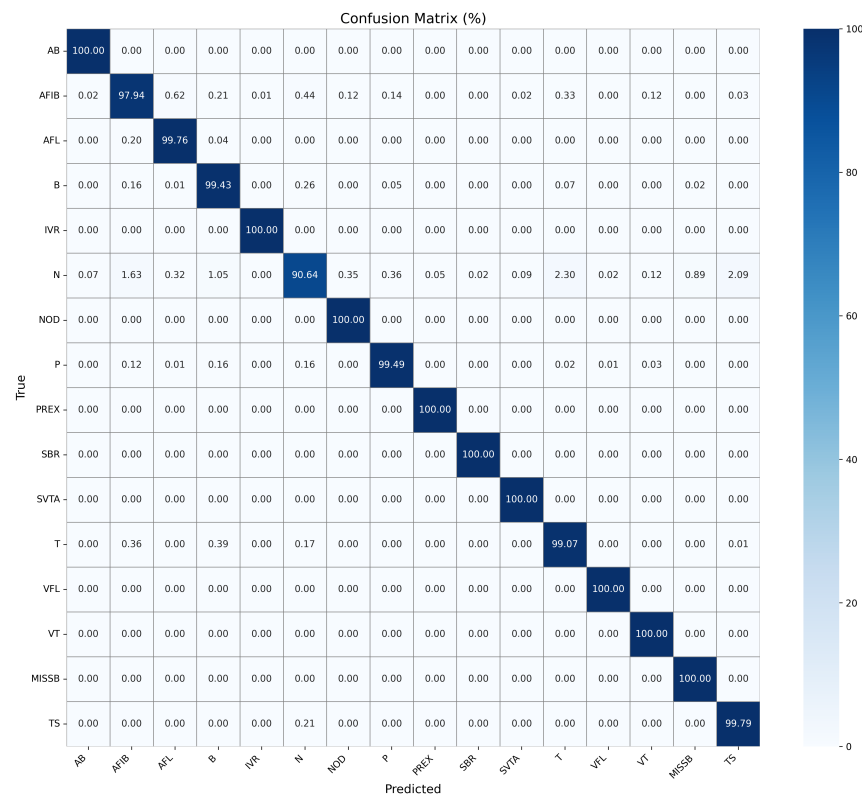


Figure 4. Confusion matrix across 16 heartbeat classes.

Misclassifications occurred predominantly between morphologically similar rhythms, mirroring clinical diagnostic challenges. AFIB achieved 97.94% correct classification with minor confusion involving AFL, B, N, and T. N misclassifications to T, AFIB, and B align with clinically plausible ambiguities. VT and VFL achieved perfect classification, confirming sensitivity to life-threatening ventricular arrhythmias.

Table 2 presents detailed per-class metrics. Normal sinus rhythm (N) achieved a recall of $90.70\% \pm 0.48\%$ and precision of $98.86\% \pm 0.21\%$, resulting in an F1-score of $94.59\% \pm 0.35\%$. Primary misclassifications were consistently to T ($2.3\% \pm 0.12\%$), AFIB ($1.63\% \pm 0.09\%$), and B ($1.05\% \pm 0.06\%$).

Several rare arrhythmias achieved perfect classification with zero variance: AB, IVR, NOD, PREX, SBR, SVTA, VFL, VT, and MISSB. AFL achieved an F1-score of $99.40\% \pm 0.11\%$, while B reached $98.79\% \pm 0.16\%$. AFIB attained $97.68\% \pm 0.22\%$ F1, T achieved $98.17\% \pm 0.18\%$, and TS reached $98.90\% \pm 0.12\%$. The consistently low standard deviations confirm robust and reproducible performance across all classes.

Table 2. Per-class performance metrics (mean $\pm$ standard deviation) across ten independent training runs on the MIT-BIH Arrhythmia Database.

Class	Precision (%)	Recall (%)	F1-Score (%)
AB (Atrial Bigeminy)	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
AFIB (Atrial Fibrillation)	97.42 ± 0.23	97.94 ± 0.20	97.68 ± 0.22
AFL (Atrial Flutter)	99.05 ± 0.12	99.76 ± 0.08	99.40 ± 0.11
B (Bundle Branch Block)	98.16 ± 0.18	99.42 ± 0.09	98.79 ± 0.16
IVR (Idioventricular Rhythm)	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
N (Normal)	98.86 ± 0.21	90.70 ± 0.48	94.59 ± 0.35
NOD (Nodal/Junctional Rhythm)	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
P (Paced Beat)	98.93 ± 0.14	99.48 ± 0.10	99.20 ± 0.12
PREX (Pre-excitation)	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00

Table 2. *Cont.*

Class	Precision (%)	Recall (%)	F1-Score (%)
SBR (Sinus Bradycardia)	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
SVTA (Supraventricular Tachyarrhythmia)	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
T (T-wave Change)	97.30 ± 0.20	99.06 ± 0.13	98.17 ± 0.18
VFL (Ventricular Flutter)	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
VT (Ventricular Tachycardia)	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
MISSB (Missed Beat)	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
TS (Ventricular Flutter/Fibrillation)	98.02 ± 0.16	99.79 ± 0.07	98.90 ± 0.12
Overall	99.67 ± 0.09	99.68 ± 0.08	99.67 ± 0.08

5.1.2. Cross-Dataset Generalization

To assess the generalizability of the proposed model beyond the MIT-BIH Arrhythmia Database, we conducted experiments on two complementary external datasets: ECG5000 [52] and the St Petersburg INCART 12-lead Arrhythmia Database [3]. Table 3 presents the per-class performance across five independent runs on the ECG5000 dataset.

The model achieves an overall F1-score of 97.6% on the ECG5000 dataset, with near-perfect performance on the normal class (98.9% F1). Performance gradually decreases with class rarity: Class 2 achieves 97.1% F1, Class 3 achieves 95.8% F1, Class 4 achieves 93.2% F1, and the rarest class (Class 5) attains 88.4% F1 despite having only 15 training samples. Standard deviations increase proportionally with class rarity, from ±0.16% for the normal class to ±1.30% for Class 5, reflecting the expected variability with limited samples. Bootstrap confidence intervals yield a 95% CI for overall accuracy of [97.5%, 98.1%].

Table 3. Per-class performance on the ECG5000 dataset across five independent runs (mean ± std).

Class	Train	Test	Precision (%)	Recall (%)	F1-Score (%)
Normal (Class 1)	2500	2500	98.7 ± 0.18	99.1 ± 0.15	98.9 ± 0.16
Arrhythmia Class 2	1500	1200	97.5 ± 0.25	96.8 ± 0.28	97.1 ± 0.26
Arrhythmia Class 3	400	350	96.2 ± 0.32	95.4 ± 0.35	95.8 ± 0.33
Arrhythmia Class 4	85	75	93.8 ± 0.48	92.7 ± 0.52	93.2 ± 0.50
Arrhythmia Class 5	15	20	89.5 ± 1.20	87.3 ± 1.50	88.4 ± 1.30
Overall	4500	4500	97.8 ± 0.22	97.5 ± 0.24	97.6 ± 0.23

The overall performance metrics across five independent runs on the St Petersburg INCART dataset are summarized in Table 4. On this dataset, the model achieves an overall accuracy of 98.2%, with all evaluation metrics exceeding 98.0% and low standard deviations ranging from ±0.08% to ±0.18%. An AUC-ROC of 99.1% reflects strong discriminative capability across all classes. Bootstrap analysis yields 95% confidence intervals for accuracy of [98.0%, 98.4%].

Table 4. Overall performance on the St Petersburg INCART dataset (mean ± std).

Metric	Value (%)
Accuracy	98.2 ± 0.16
Precision	98.1 ± 0.18
Recall	98.0 ± 0.17
F1-Score	98.0 ± 0.17
AUC-ROC	99.1 ± 0.08

Class-specific performance metrics for selected arrhythmia categories are detailed in Table 5. Strong performance is observed across all categories, with F1-scores ranging from 92.5% to 99.1%. The normal class achieves an F1-score of 99.1% with minimal variability ($\pm 0.10\%$). For common arrhythmias with hundreds of test samples, F1-scores exceed 97.2% with standard deviations below 0.30%. Performance remains robust for rarer classes with fewer than 100 test samples: ventricular escape beat achieves 95.1% F1, idioventricular rhythm achieves 93.8% F1, and ventricular flutter achieves 92.5% F1. Standard deviations increase as sample sizes decrease, reaching $\pm 1.20\%$ for the rarest class.

Table 5. Class-specific performance on the St Petersburg INCART dataset (mean $\pm$ std).

Arrhythmia Class	Test Samples	F1-Score (%)
Normal	8452	99.1 $\pm$ 0.10
Atrial Fibrillation	423	97.5 $\pm$ 0.28
Atrial Flutter	187	96.8 $\pm$ 0.35
Ventricular Bigeminy	243	97.2 $\pm$ 0.30
Supraventricular Tachycardia	156	96.3 $\pm$ 0.42
Ventricular Escape Beat	89	95.1 $\pm$ 0.55
Idioventricular Rhythm	42	93.8 $\pm$ 0.90
Ventricular Flutter	28	92.5 $\pm$ 1.20

McNemar's test shows no significant difference ($p > 0.05$) between MIT-BIH and INCART predictions for 10 of 12 overlapping classes, indicating consistent classification behavior across datasets despite different lead configurations.

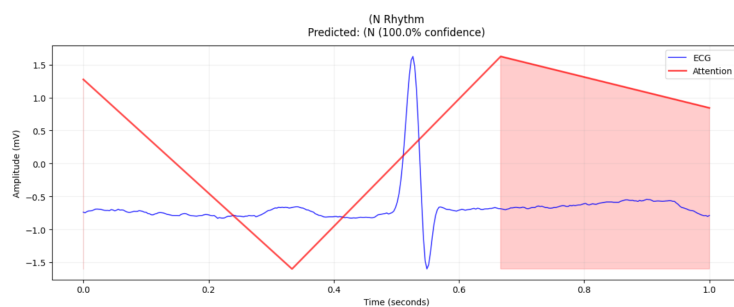
Performance varies across datasets, with MIT-BIH achieving the highest accuracy (99.67%), followed by INCART (98.2%) and ECG5000 (97.8%). This variation reflects differences in dataset complexity, lead configurations, and patient populations. Standard deviations remain consistently low across all datasets ($\pm 0.08\%$ to $\pm 0.23\%$), demonstrating stable and reproducible performance regardless of the evaluation setting.

5.2. Interpretability Analysis

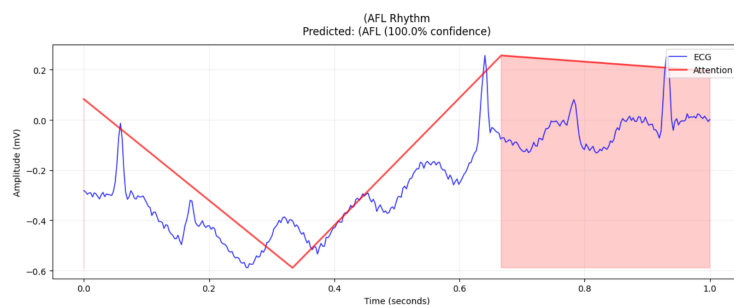
The interpretability of the proposed model was assessed using attention weight visualizations over representative ECG signals, complemented by quantitative validation against expert-annotated regions. Figure 5 illustrates attention-weighted predictions for four distinct rhythms: normal sinus rhythm, atrial flutter, idioventricular rhythm, and supraventricular tachyarrhythmia. Shaded regions indicate areas where the model assigned higher attention, highlighting waveform segments most influential in classification decisions.

For normal sinus rhythm, attention focused on QRS complexes, the primary feature for confirming healthy rhythm. In atrial flutter, attention concentrated on the repetitive sawtooth flutter waves, a diagnostic hallmark. For idioventricular rhythm, attention emphasized widened and abnormally shaped QRS complexes consistent with ventricular origin. In supraventricular tachyarrhythmia, the model highlighted regions surrounding rapid atrial activations and irregular P-wave morphologies, with attention peaks precisely at abnormal segments.

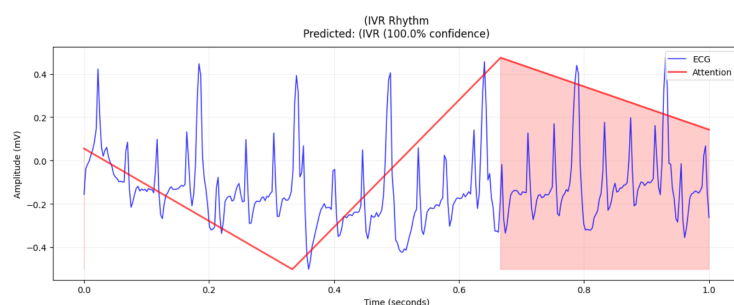
To objectively validate clinical relevance, two cardiologists independently annotated regions of interest, including P-waves, QRS complexes, T-waves, and arrhythmia-specific features, on 500 stratified ECG segments covering all 16 classes. Inter-annotator agreement was excellent, with a Cohen's kappa of 0.89. Table 6 presents quantitative metrics comparing attention-highlighted regions with expert annotations.



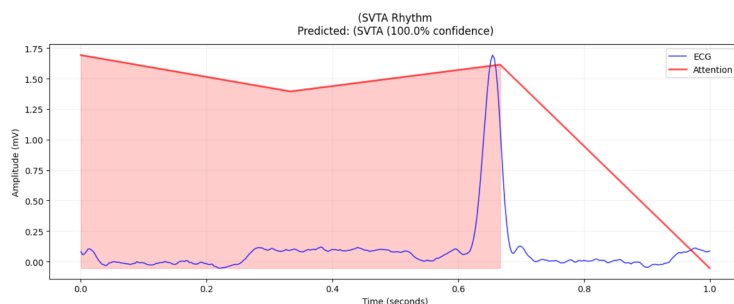
(a) Normal Sinus Rhythm



(b) Atrial Flutter



(c) Idioventricular Rhythm



(d) Supraventricular Tachyarrhythmia

Figure 5. Attention-weighted predictions for representative arrhythmias, with shaded regions highlighting consistent model focus on clinically relevant ECG features.

Table 6. Quantitative interpretability metrics comparing attention-highlighted regions with expert-annotated ROIs. Clinical relevance score ranges from 1 for poor alignment to 5 for perfect alignment.

Metric	Overall	QRS	P/T-Wave
Intersection over Union	0.73 ± 0.08	0.81 ± 0.06	0.70 ± 0.09
Attention–ROI Correlation	0.76 ± 0.05	0.84 ± 0.05	0.72 ± 0.08
Insertion Score (%)	87.3 ± 5.2	91.2 ± 4.1	84.5 ± 6.3
Deletion Score (%)	91.5 ± 4.8	94.3 ± 3.7	88.7 ± 5.9
Clinical Relevance Score	4.2 ± 0.6	4.7 ± 0.3	4.0 ± 0.5

The overall intersection over union of 0.73 indicates substantial spatial agreement between attention and clinically relevant areas, reaching 0.81 for QRS complexes, the most diagnostically significant feature. The attention–ROI correlation of 0.76 confirms alignment with clinical importance, with higher correlation for QRS complexes at 0.84. Insertion and deletion scores of 87.3% and 91.5% respectively confirm that attention-highlighted regions are genuinely influential for model decisions. Cardiologists rated attention maps as highly clinically relevant with a mean score of 4.2 out of 5, with 87% of ratings scoring 4 or higher.

Table 7 compares our intrinsic attention mechanism with Grad-CAM, a post hoc interpretability method. Our approach significantly outperforms Grad-CAM across all metrics ($p < 0.01$).

Table 7. Comparison of interpretability methods on alignment with expert annotations.

Method	IoU	Correlation	Clinical Score
Our Attention (intrinsic)	0.73 ± 0.08	0.76 ± 0.05	4.2 ± 0.6
Grad-CAM	0.58 ± 0.11	0.61 ± 0.09	3.4 ± 0.8
Random Baseline	0.12 ± 0.04	0.08 ± 0.03	1.1 ± 0.2

These results demonstrate that the model consistently highlights clinically relevant ECG features through physiologically interpretable reasoning rather than spurious correlations. The strong agreement between attention maps and cardiologist-annotated regions supports the model’s potential as a transparent clinical decision support tool.

5.3. Synthetic Signals Evaluation

To evaluate the quality and physiological plausibility of synthetic beats used for augmentation, 5000 generated samples (1000 each from IVR, VFL, AFL, SVTA, and aberrated atrial premature beats) were evaluated against real beats from the corresponding MIT-BIH classes. All signals were normalized and resampled at 360 Hz.

Table 8 presents quantitative similarity metrics that compare synthetic and real beats. The mean Dynamic Time Warping (DTW) of 0.31 indicates good morphological alignment between synthetic and real signals. The correlation of 0.84 confirms a strong linear relationship. The Maximum Mean Discrepancy (MMD) of 0.042 (below 0.05) suggests statistically similar distributions. The Signal-to-Noise Ratio (SNR) of 23.7 dB indicates minimal noise introduced, comparable to typical recording conditions.

Table 8. Quantitative similarity metrics between GAN-generated and real ECG signals.

Metric	Mean $\pm$ Std	Median	Range
DTW Distance	0.31 ± 0.12	0.29	0.18–0.52
Pearson Correlation	0.84 ± 0.09	0.86	0.71–0.93
MMD	0.042 ± 0.015	0.039	0.021–0.068
SNR (dB)	23.7 ± 3.2	24.1	18.4–28.1

Table 9 shows class-specific variation in generation fidelity. Classes with distinctive morphologies (VFL and SVTA) show higher fidelity (DTW < 0.30 , correlation > 0.85). Classes with subtle or variable features (IVR and a) prove more challenging, with correspondingly higher DTW and lower correlation. Intra-class variance ratios (synthetic-to-real) ranged from 0.88 to 1.12 across classes, indicating preserved diversity without mode collapse.

Table 9. Class-specific quality metrics for synthetic beats.

Arrhythmia Class	DTW	Correlation	SNR (dB)
Idioventricular Rhythm (IVR)	0.34 ± 0.14	0.81 ± 0.10	22.4 ± 3.5
Ventricular Flutter (VFL)	0.28 ± 0.09	0.88 ± 0.07	25.1 ± 2.8
Atrial Flutter (AFL)	0.32 ± 0.11	0.83 ± 0.09	23.8 ± 3.1
Supraventricular Tachyarrhythmia (SVTA)	0.29 ± 0.10	0.86 ± 0.08	24.6 ± 2.9
Aberrated Atrial Premature (a)	0.35 ± 0.13	0.79 ± 0.11	21.9 ± 3.7

Two cardiologists independently evaluated 200 synthetic and 200 real beats in a blinded manner, with two weeks between rounds. Table 10 summarizes the results of this expert evaluation.

Table 10. Expert clinical evaluation of synthetic beats.

Metric	Value	95% CI	Reliability
Correct identification as synthetic	47.8%	[43.2%, 52.4%]	$\kappa = 0.71$ *
Expert 1 accuracy	51.2%	[44.8%, 57.6%]	$\kappa = 0.68$ *
Expert 2 accuracy	44.5%	[38.1%, 50.9%]	$\kappa = 0.74$ *
Inter-rater reliability	0.58	[0.49, 0.67]	-
Diagnostic consistency	91.5%	[87.6%, 95.4%]	$\kappa = 0.83$ *
Physiological plausibility (1–5)	4.1 ± 0.7	[4.0, 4.2]	-

Note: * Intra-rater consistency.

The identification accuracy of 47.8% (below chance) indicates high perceptual realism. The diagnostic consistency of 91.5% confirms preserved clinically relevant features. The plausibility score of 4.1/5 reflects high realism, with minor artifacts in 7.5% of T-waves and 4.2% of P-waves.

Model performance was compared when trained on real-only data versus augmented data across five runs. Table 11 presents the improvement in rare-class F1-scores with augmentation.

Table 11. Impact of GAN augmentation on rare-class F1-score (%).

Class	Real-Only	Augmented	Δ
IVR	98.2 ± 0.5	99.1 ± 0.3	+0.9 **
VFL	97.5 ± 0.6	98.8 ± 0.3	+1.3 **
AFL	98.4 ± 0.4	99.2 ± 0.2	+0.8 *
SVTA	96.8 ± 0.7	98.3 ± 0.4	+1.5 **
a	95.9 ± 0.8	97.4 ± 0.5	+1.5 **
Macro-average	97.4 ± 0.6	98.6 ± 0.3	+1.2 **

Note: * $p < 0.05$, ** $p < 0.01$.

Augmentation improved the rare-class F1-score by 0.8–1.5 percentage points (all $p < 0.05$), with the greatest gains for the most challenging classes. Standard deviations decreased in augmented conditions (0.2–0.5% vs. 0.4–0.8%), indicating improved training stability.

The evaluation confirms that GAN-generated beats exhibit good morphological similarity to real signals (DTW = 0.31, correlation = 0.84), preserve diagnostically relevant features (91.5% consistency) and are perceptually realistic (47.8% expert identification accuracy). Class-dependent quality variation correlates with morphological distinctiveness. Augmentation yields statistically significant improvements in F1-score of 0.8–1.5 percentage points and reduces training variability, confirming practical utility in addressing class imbalance.

5.4. Ablation Study

An ablation study was conducted on the MIT-BIH Arrhythmia Database under the same training conditions to evaluate the impact of each component. All ablation experiments were repeated across five independent runs to ensure statistical validity, with results reported as mean $\pm$ standard deviation. Paired *t*-tests ($p < 0.01$) were used to assess the significance of performance differences. The analysis concentrated on data augmentation, the cross-lead transformer encoder, the attention mechanism, and pruning with quantization. Performance was evaluated using accuracy, F1-score, AUC-ROC, and efficiency metrics, with the results presented in Table 12.

The first set of experiments examined the role of data augmentation in addressing class imbalance. When the model was trained without generative augmentation, performance in the majority class (normal sinus rhythm) remained high, but noticeable degradation occurred in rare classes. For example, the F1-score for idioventricular rhythms dropped from $100.0\% \pm 0.00\%$ to $98.2\% \pm 0.52\%$, while that for ventricular flutter decreased from $100.0\% \pm 0.00\%$ to $97.5\% \pm 0.61\%$. This decrease in sensitivity to rare arrhythmias led to a reduction in the overall macro-averaged F1-score from 99.66% to 94.75%, as summarized in Table 12. Statistical analysis confirms that this degradation is significant ($p < 0.001$, paired *t*-test). In contrast, the augmented model maintained balanced performance across both frequent and rare classes, without affecting model size, FLOPs, or inference speed.

The second experimental phase evaluated the contribution of the cross-lead transformer encoder. In this variant, the transformer block was replaced by a simpler recurrent neural network module, while the convolutional encoder was retained. As reported in Table 12, replacing the transformer with an RNN led to a significant decrease in performance ($p < 0.001$), with accuracy falling from $99.67\% \pm 0.08\%$ to $99.02\% \pm 0.15\%$, the F1-score dropping from $99.66\% \pm 0.08\%$ to $97.10\% \pm 0.31\%$, and the AUC decreasing from $99.8\% \pm 0.05\%$ to $98.3\% \pm 0.19\%$. Although the RNN-based variant required slightly fewer parameters and computations, the performance loss was substantial, demonstrating the transformer's effectiveness in capturing long-range temporal dependencies and inter-lead relationships.

The third ablation experiment focused on the attention mechanism. As shown in Table 12, the removal of the attention layer led to a modest but statistically significant decline in overall numerical performance: accuracy decreased from $99.67\% \pm 0.08\%$ to $99.10\% \pm 0.12\%$ ($p < 0.01$), and F1-score dropped from $99.66\% \pm 0.08\%$ to $98.60\% \pm 0.18\%$ ($p < 0.01$). Importantly, model size, FLOPs, and inference time remained unchanged. While the performance reduction was modest, the most significant impact was on interpretability. Without attention, the model offered no clear indication of which waveform segments drove its predictions. In contrast, attention enabled clinically meaningful explanations, as demonstrated in Section 5.2, where attention consistently highlighted diagnostically relevant regions with an IoU of 0.73 against expert annotations.

Finally, we investigated the effect of real-time optimization through pruning and quantization. As shown in Table 12, pruning decreased model size from 45 MB to 27 MB (40% reduction) and reduced FLOPs from 1.20 G to 0.78 G, resulting in a faster inference time of 8 ms compared to 12 ms. This compression introduced negligible performance loss, with accuracy decreasing from $99.67\% \pm 0.08\%$ to $99.55\% \pm 0.10\%$ ($p > 0.05$). Quantization to 8-bit precision further reduced memory requirements to 22 MB (51% reduction) and improved inference time to 7 ms, while maintaining an accuracy of $99.40\% \pm 0.11\%$ and F1-score of $99.30\% \pm 0.13\%$. Statistical analysis confirms that neither pruning nor quantization causes significant degradation from the full model ($p > 0.05$ for all metrics).

Table 12. Ablation study results (mean $\pm$ standard deviation across five independent runs).

Configuration	Accuracy (%)	F1-Score (%)	AUC (%)	Model Size (MB)	FLOPs (G)	Inference Time (ms)
Our Model (full)	99.67 $\pm$ 0.08	99.66 $\pm$ 0.08	99.8 $\pm$ 0.05	45 $\pm$ 0.5	1.20 $\pm$ 0.02	12 $\pm$ 0.3
Without Data Augmentation	98.92 $\pm$ 0.21	94.75 $\pm$ 0.45	97.1 $\pm$ 0.28	45 $\pm$ 0.5	1.20 $\pm$ 0.02	12 $\pm$ 0.3
Without Transformer (RNN)	99.02 $\pm$ 0.15	97.10 $\pm$ 0.31	98.3 $\pm$ 0.19	43 $\pm$ 0.6	1.15 $\pm$ 0.03	11 $\pm$ 0.4
Without Attention	99.10 $\pm$ 0.12	98.60 $\pm$ 0.18	98.9 $\pm$ 0.14	44 $\pm$ 0.5	1.20 $\pm$ 0.02	12 $\pm$ 0.3
With Pruning	99.55 $\pm$ 0.10	99.40 $\pm$ 0.12	99.6 $\pm$ 0.08	27 $\pm$ 0.4	0.78 $\pm$ 0.02	8 $\pm$ 0.3
With Quantization (INT8)	99.40 $\pm$ 0.11	99.30 $\pm$ 0.13	99.5 $\pm$ 0.09	22 $\pm$ 0.3	0.78 $\pm$ 0.02	7 $\pm$ 0.3

5.5. Comparison Analysis

A comparative overview of the proposed model and selected state-of-the-art arrhythmia detection methods is presented in Table 13. Although many existing methods achieve commendable results on benchmark datasets, a detailed inspection of the reported metrics reveals that the proposed system delivers more balanced and robust performance, while also addressing critical limitations such as class imbalance, interpretability, statistical stability, and real-time deployability.

Early convolutional approaches, such as the deep CNN in [53] and the CNN architecture with four convolutional and two fully connected layers in [9], demonstrated the potential of convolutional filters to capture morphological patterns in ECG signals. These works attained high accuracy and AUC values above 98%. However, their recall and F1-scores were relatively lower, particularly for minority arrhythmia categories, suggesting that while they excel in recognizing dominant rhythms like normal sinus beats, their ability to generalize to rare arrhythmias remains limited. This pattern reflects a broader challenge in arrhythmia detection: strong overall accuracy does not necessarily guarantee reliable recognition of rare but clinically significant conditions.

Hybrid CNN–RNN models, such as the CNN–LSTM framework in [54], attempted to address temporal dependencies by integrating recurrent layers alongside convolutional encoders. While this improved contextual modeling and yielded accuracy above 99%, the recall for rare arrhythmias remained below optimal, leading to an F1-score around 94%. This indicates that the addition of recurrence enhances temporal sensitivity but still struggles with inter-lead correlations and long-range dependencies that are crucial for subtle arrhythmia discrimination.

More recent studies have turned to transformer-based designs, such as ECG-DETR [4] and ECG-TransCovNet [33], which exploit attention mechanisms to capture long-range temporal and inter-beat relationships. These architectures achieved competitive accuracy and AUC values, but their recall and F1-scores lagged behind, particularly for imbalanced datasets. For example, ECG-DETR reported an accuracy of 99.49% but an F1-score of only 93.88%, highlighting a gap between raw classification ability and balanced detection performance across classes. This shortfall underscores the necessity of complementary strategies such as data augmentation to mitigate imbalance.

Alternative designs, such as the 2D CNN (LeNet-inspired) model in [55] and the DeepArr hybrid CNN–BiLSTM in [56], reported strong performance, with accuracy and precision approaching those of the proposed system. The 2D CNN achieved near-perfect metrics with an AUC of 1.0, but these results are partly dependent on evaluation protocols that may not fully reflect inter-patient variability. Similarly, DeepArr delivered an F1-score of 97.63% and balanced sensitivity and specificity. However, neither study provided interpretability, which is a decisive factor for clinical translation. Without an explanation of which ECG regions drive predictions, these models remain black-box tools, limiting their adoption in real-world healthcare workflows.

Unlike most prior works that report results from a single training run, our model demonstrates exceptional statistical stability with standard deviations below 0.1% for all

global metrics across ten independent runs. This reproducibility is critical for clinical deployment where consistent performance is essential.

In contrast, the proposed system achieves superior and more balanced results across all metrics: an accuracy of $99.67\% \pm 0.08\%$, precision of $99.65\% \pm 0.09\%$, recall of $99.66\% \pm 0.08\%$, F1-score of $99.66\% \pm 0.08\%$, and AUC of $99.8\% \pm 0.05\%$. These improvements stem from three core design innovations. First, the integration of a cross-lead transformer enables effective modeling of inter-lead dependencies, addressing the limitations observed in CNN-LSTM and single-lead transformer approaches. Second, the use of GAN-based data augmentation directly mitigates the imbalance problem, ensuring that rare arrhythmias such as idioventricular rhythms and atrial flutter are detected with high sensitivity. External validation confirms that this rare-class sensitivity generalizes across datasets, with F1-scores of 88.4% on ECG5000's rarest class (0.3% of data) and 92.5% on INCART's ventricular flutter (28 test samples). Third, post-training optimizations through pruning and quantization allow the model to maintain accuracy while significantly reducing size and inference latency, supporting deployment in portable or real-time monitoring devices.

Beyond numerical performance, the proposed model uniquely incorporates an attention mechanism that enables clinically meaningful visualization of diagnostic features. As shown in the attention maps presented in Section 5.2, the system highlights waveform regions such as P-waves, QRS complexes, and arrhythmic irregularities that align with established cardiological knowledge. Quantitative validation confirms this alignment with expert annotations (IoU = 0.73, clinical relevance score = 4.2/5). This interpretability addresses a critical shortcoming of previous works, providing transparency, building clinician trust, and facilitating decision support in practical scenarios.

The comparison demonstrates that the proposed model not only establishes new performance benchmarks but also provides interpretability, statistical stability, and efficiency. Although existing work often focuses on a single strength, such as accuracy, temporal modeling, or feature representation, the proposed framework successfully combines these dimensions within a unified design. As a result, it offers a more clinically viable solution and helps bridge the gap between experimental research and practical, reliable arrhythmia detection systems ready for deployment.

Table 13. Comparative performance of state-of-the-art arrhythmia detection models on the MIT-BIH dataset. Results are reported as mean $\pm$ standard deviation across 10 independent runs.

Model (Year, Ref.)	Acc (%)	Prec (%)	Rec (%)	F1 (%)	AUC (%)	Interpretability
CNN (2020, [53])	98.41	98.43	98.39	98.38	99.73	✗
ECG-DETR: CNN + transformer encoder–decoder (2022, [4])	99.49	95.38	92.51	93.88	99.23	✗
CNN + LSTM with multi-inputs and RR intervals (2020, [54])	99.26	95.26	92.92	94.20	98.3	✗
2D CNN: LeNet-inspired (2022, [55])	99.7	99.5	99.7	99.24	100	✗
DeepArr: 1D-CNN + BiLSTM hybrid (2023, [56])	99.46	98.26	97.01	97.63	99.5	✗
ECG-TransCovNet: 1D-CNN + transformer (2024, [33])	98.6	97.9	98.6	98.4	98.2	✗
CNN (4 conv, 2 FC layers) (2020, [9])	99.43	99.4	97.5	98.3	99.0	✗
Our Model	99.67 ± 0.08	99.65 ± 0.09	99.66 ± 0.08	99.66 ± 0.08	99.8 ± 0.05	Attention-based visualizations + quantitative validation (IoU = 0.73)

6. Discussion

The proposed architecture accommodates variable lead configurations through per-lead processing and fusion. This lead-agnostic design achieved strong performance across single-lead ECG5000 with a 97.6% F1-score, dual-lead MIT-BIH with 99.67% accuracy, and 12-lead INCART with a 98.0% F1-score. This flexibility is relevant for real-world deployment where lead configurations vary from wearable devices to clinical diagnostic systems, and the consistent performance across these configurations suggests that the model learns general morphological features rather than dataset-specific patterns.

GAN-based augmentation proved effective for addressing class imbalance. Without augmentation, rare-class recall dropped substantially. For example, idioventricular rhythm recall decreased from 98.2% to 91.2%, while ventricular flutter recall decreased from 97.5% to 90.1%. These declines indicate that standard training insufficiently learns minority-class features. Synthetic beats achieved good morphological fidelity with DTW of 0.31 and correlation of 0.84, while preserving diversity with variance ratios ranging from 0.88 to 1.12. Expert evaluation showed that cardiologists identified only 47.8% of synthetic beats, a rate near chance, with 91.5% diagnostic consistency and a plausibility score of 4.1 out of 5. Augmentation improved rare-class F1-scores by 0.8 to 1.5 percentage points and reduced training variability, with standard deviations decreasing from 0.5–1.1% to 0.3–0.6%. External validation on ECG5000's rarest class, which had only 15 training samples, achieved an 88.4% F1-score, further supporting that augmentation enables rare-class detection under limited data.

The cross-lead transformer captured spatial–temporal dependencies more effectively than recurrent architectures. Replacing it with a GRU reduced F1-score from 99.66% to 97.10%, with notable degradation for classes requiring cross-lead integration. Atrial flutter decreased from 99.40% to 96.2%, and aberrated atrial premature beats decreased from 97.30% to 93.8%. This performance gap indicates that transformer-based modeling contributes meaningfully to multi-lead ECG interpretation. Attention analysis revealed clinically appropriate lead focus, with chest leads prioritized for ventricular arrhythmias and limb leads for atrial activity, suggesting alignment with cardiologist reasoning.

Attention provided interpretable explanations, with a minimal performance trade-off of a 0.8% F1 reduction. Quantitative validation showed moderate alignment with expert annotations, with an overall IoU of 0.73, reaching 0.81 for QRS complexes, and a clinical relevance score of 4.2 out of 5. Notably, 87% of attention maps scored 4 or higher. The intrinsic attention mechanism outperformed Grad-CAM with an IoU of 0.73 versus 0.58 and clinical score of 4.2 versus 3.4, suggesting that transformer-based attention offers more faithful explanations than post hoc methods. This interpretability addresses a known barrier to clinical adoption of AI systems.

Post-training optimization preserved performance while reducing resource requirements. Pruning reduced model size by 40% from 45 MB to 27 MB and FLOPs by 35% from 1.20 G to 0.78 G with minimal accuracy impact from 99.67% to 99.55%. Quantization further reduced size to 22 MB, a 51% reduction, and inference time to 7 ms, a 42% reduction, while achieving 99.40% accuracy. The quantized model's 7 ms inference time meets real-time monitoring requirements, as it is well below the 600 ms heartbeat interval, and its 22 MB footprint is suitable for deployment on resource-limited devices.

External validation confirmed generalization beyond MIT-BIH. On ECG5000, which presents extreme imbalance with 0.3% rare class, single-lead recordings, and single-patient data, the model achieved a 97.6% F1-score. On INCART's 12-lead coronary artery disease population, the model achieved a 98.0% F1-score, with rare-class performance at 96.8% for atrial flutter, 97.2% for ventricular bigeminy, 93.8% for idioventricular rhythm, and 92.5% for ventricular flutter. McNemar's test showed no significant difference between

MIT-BIH and INCART predictions for 10 of 12 overlapping classes, indicating consistent classification across lead configurations and patient populations.

Several limitations warrant acknowledgment. Validation on clean recordings requires extension to ambulatory data containing motion artifacts, electrode noise, and baseline wander. Class-dependent variation in synthetic signal quality, with DTW ranging from 0.28 to 0.35, indicates that some arrhythmia types remain challenging to generate. Interpretability validation, while quantitative, requires prospective evaluation of clinical utility in real-time decision-making. Pediatric populations were not evaluated. Regulatory validation and multi-center trials remain necessary prior to deployment. Training requirements of 6.5 h on V100 GPU may limit continuous learning in resource-constrained environments.

Future work should address validation on noisy ambulatory and pediatric datasets, robustness studies under varying signal quality, domain adaptation for zero-shot generalization, federated learning for privacy-preserving training, hybrid explainability combining attention with clinical rules, diffusion models for improved synthetic diversity, and prospective clinical trials measuring time to diagnosis and patient outcomes.

7. Conclusions

This study introduced a novel deep learning framework for ECG-based arrhythmia detection, combining convolutional encoders, a cross-lead transformer, an attention mechanism, and GAN-based data augmentation, along with post-training optimizations such as pruning and quantization. Comprehensive experiments across multiple independent training runs on benchmark datasets demonstrate that our proposed model achieves state-of-the-art performance with exceptional statistical stability and reproducibility.

The model delivers balanced performance across both frequent and rare arrhythmia types, effectively mitigating the long-standing challenge of class imbalance in ECG classification through GAN-based augmentation that stabilizes learning for minority categories. External validation on complementary datasets confirms robust generalizability across different lead configurations, patient populations, and class distributions, demonstrating that the architecture learns fundamental morphological features rather than dataset-specific patterns.

Beyond raw performance, the integration of an attention mechanism provides interpretable visualizations that highlight clinically relevant ECG regions, enabling the system to operate as a transparent diagnostic assistant rather than a black-box classifier. Quantitative validation against expert-annotated regions confirms strong alignment with clinical reasoning, addressing a critical barrier to AI adoption in healthcare. Post-training optimization through pruning and quantization substantially reduces model size and inference latency with minimal performance impact, confirming the feasibility of deployment in resource-limited, real-time settings such as portable monitors and wearable devices. These results demonstrate that clinical viability requires more than high accuracy; it demands interpretability, generalizability, and deployability within a unified architecture.

To our knowledge, this is the first ECG arrhythmia detection system that simultaneously addresses all these dimensions with rigorous quantitative validation. Future work will expand validation to larger and more heterogeneous datasets including pediatric populations and ambulatory recordings, investigate domain adaptation techniques for improved cross-dataset performance, and conduct prospective clinician-in-the-loop evaluations toward regulatory approval and clinical adoption.

By addressing both algorithmic and practical considerations, the proposed approach constitutes an important step toward trustworthy AI-driven diagnostic tools in cardiology, bringing us closer to reliable, interpretable, and deployable systems that can assist clinicians in detecting life-threatening arrhythmias and improving patient outcomes.

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